

Grade 12 Chemistry Unit 3 the Physical state of matters

At the end of this unit, you will be able to:

- **discuss the kinetic molecular theory and properties of the three physical states of matter**
- **explain the behavior of gases by using the variables volume, temperature, pressure and number of moles**
- **distinguish these terms: ideal gas, diffusion, evaporation, boiling, condensation, vapour pressure, boiling point, molar heat of vaporization, molar heat of condensation, melting, fusion, sublimation, melting point, freezing point, molar heat of fusion, and molar heat of solidification**
- **analyse the gas laws**
- **develop skills in solving problems to which the gas laws apply**
- **perform activities to illustrate gas laws**
- **carryout experiments to determine the boiling points of liquids**
- **demonstrate an experiment to show phase changes.**

3.1 INTRODUCTION

At the end of this section, you will be able to:

- describe the physical states of matter
- give examples for each of the three physical states of matter
- compare and contrast the three physical states of matter.

3.2 THE KINETIC THEORY OF MATTER

At the end of this section, you will be able to:

- state the kinetic theory of matter
- explain the properties of matter in terms of the kinetic theory.

3.3 THE GASEOUS STATE

At the end of this section, you will be able to:

- explain the assumptions of the kinetic molecular theory of gases
- describe the properties of gases using the kinetic molecular theory
- describe the behavior of gases by using the variables V (volume), T (temperature), P (pressure) and n (number of moles)
- state Boyle's law
- perform an activity to show changes in volume and pressure of gases to illustrates Boyle's law
- apply Boyle's law in solving problems
- *state Charles' law*
- *conduct an activity to show changes in volume and temperature of gases to illustrate Charles' law*
- *apply Charles' law in solving problems*
- *state Gay-Lussac's law*
- *use Gay-Lussac's law in solving problems*

- *derive the combined gas law equation from Boyle's law, Charles' law, and Gay-Lussac's law*
- *use the combined gas law to calculate changes in volume, pressure or temperature*
- *define an ideal gas*
- *derive an ideal gas equation from Boyle's law, Charles' law and Avogadro's law*
- *compare the nature of real gases with ideal gases*
- *solve problems related to the ideal gas equation*
- *define diffusion*
- *state Graham's law of diffusion*
- *carry out an activity to compare the rate of diffusion of two different gases*

3.4 THE LIQUID STATE

At the end of this section, you will be able to:

- explain the terms: evaporation, boiling, condensation, vapor pressure;
- boiling point, molar heat of vaporization and molar heat of condensation
- carry out an activity to demonstrate the concept of vapor pressure
- carry out an activity to determine the boiling points of water and ethanol.

3.5 SOLID STATE

At the end of this section, you will be able to:

- describe phase changes
- explain the terms melting, fusion, sublimation, melting point, freezing point, molar heat of solidification
- explain temperature changes associated with phase changes
- determine melting point of ice
- demonstrate an experiment to show the phase changes from ice to liquid water and then to water vapor.